SEMESTER 5 – WEEK 3 – DESCRIBING SCIENTIFIC PROCESSES

1. Reading Comprehension

The Power of Process Thinking in Science

Modern science is built on a clear understanding of processes—how things change, react, or develop over time. Whether we are observing the water cycle, analyzing chemical reactions, or studying photosynthesis, we rely on a logical sequence of actions and results. Process thinking allows scientists to explain not only *what* happens but also *how* and *why* it happens.

Take the process of **photosynthesis**, for example. It is not a random event but a chain of carefully ordered steps. First, light energy from the sun is absorbed by chlorophyll in the leaves. Then, this energy is converted into chemical energy that allows the plant to transform carbon dioxide and water into glucose and oxygen. The overall process is essential for life on Earth because it produces the oxygen that humans and animals breathe.

In laboratories, process-based observation is equally important. When chemists design an experiment, they must describe every stage in detail: the materials used, the procedure followed, and the results observed. Scientific accuracy depends on being able to repeat the same process and obtain the same outcome. This is why lab reports use **the passive voice**, emphasizing the process itself rather than the person performing it. For instance, instead of writing *"We heated the solution,"* scientists write *"The solution was heated to 80°C."* This focuses attention on the action and maintains objectivity.

Process thinking is not limited to natural sciences; engineers use it to design machines, and computer scientists apply it in programming algorithms. Understanding and describing processes train the mind to be systematic, patient, and precise. As the saying goes, *"Science is not only about discovery—it is about understanding how discovery happens."*

A. General Understanding

1. What is the main idea of the text?
2. Why is describing a process important in science?

B. True / False / Not Mentioned

Statement	T / F / NM
Photosynthesis converts oxygen into carbon dioxide.	F
The passive voice is used to make scientific writing more objective.	T
Engineers do not use process thinking.	F
The author believes process thinking improves logical skills.	NM

C. Multiple Choice

1. The main benefit of process thinking is:
   a) Memorizing facts
   b) Understanding cause and effect
   c) Writing faster reports
   d) Avoiding experiments

2. In scientific writing, the passive voice is preferred because:
   a) It makes sentences shorter
   b) It focuses on the process, not the person
   c) It sounds more emotional
   d) It avoids repetition

D. Inference & Short Answer

1. Why is photosynthesis described as "essential for life on Earth"?
2. What qualities does process thinking help develop in scientists?
3. Find one example of the **passive voice** in the text and explain its function.

2. Vocabulary Exercises

A. Match the Word to Its Meaning

Word	Meaning
sequence	a) the result or effect of an action
convert	b) order in which events happen
outcome	c) to change something from one form to another
accuracy	d) being correct or exact
emphasize	e) to give special importance to something

B. Word Formation

Verb	Noun	Adjective
analyze	_____	analytical
react		
describe	_____	descriptive
observe		

C. Use the Words in Context

Complete the sentences using the correct form of the words above.

1. Every experiment must include an accurate _____ of the steps.
2. The _____ of the test was different from what we expected.
3. Scientists often _____ data to identify patterns.
4. Temperature can influence how fast chemicals _____ with each other.

3. Grammar – Sequencing and Passive Voice

A. Explanation

1. Sequencing Expressions are used to describe the order of actions in a process:

First, Then, After that, Next, Finally, At the end, Once

Example: *First, the solution is mixed with water. Then, it is heated to 60°C.*

2. Passive Voice
Used when the focus is on the action, not the actor.
Form: **be + past participle**

- *The sample is analyzed in the laboratory.*
- *The mixture was heated to 100°C.*

Why use it?
In scientific writing, the process is more important than who performs it.

B. Practice 1 – Rewrite in the Passive Voice

1. The students recorded the temperature every five minutes.
   → _____
2. We cleaned the test tubes after the experiment.
   → _____
3. They will publish the results in the university journal.
   → _____
4. Scientists discovered a new element.
   → _____

C. Practice 2 – Order the Steps

Put the sentences in the correct order using sequencing words.

1. The plant produces glucose.
2. Chlorophyll absorbs sunlight.
3. Carbon dioxide and water enter the leaves.
4. Oxygen is released into the air.

→ **Correct sequence:** _____

D. Practice 3 – Complete Sentences

Use the words: *first, then, after that, finally*

1. _____, the materials are collected.
2. _____, the liquid is poured into the beaker.
3. _____, it is heated slowly.
4. _____, the temperature is recorded.

4. Speaking:

In groups of three:

1. Choose a simple scientific process (e.g., water cycle, digestion, distillation).
2. Prepare a short oral explanation (2–3 minutes) describing it step by step using **sequencing words** and **passive voice**.
3. One student presents; others ask clarification questions.

5. Writing:

Write a paragraph (180–200 words) describing one of the following:

- *The process of water purification*
- *The life cycle of a plant*
- *How a scientific experiment is conducted in your field*

Instructions:

- Use at least **five sequencing expressions** (first, next, etc.)
- Include at least **five passive voice structures**
- Use vocabulary from this week (e.g., sequence, convert, outcome, accuracy)